DP Biology – D1.3

Mutation

- How do mutations in genes arise?
- What effects do gene mutations have?

SL statements D1.3.1–7. Textbook pp. 637–642. Gene editing is at HL.
Image: a barley field. Many crop varieties carry useful mutations. Public Domain Images.



Objectives for each class

Class	Statements	By the end of the class, you can...
1	D1.3.1–3	Distinguish substitutions, insertions and deletions, and explain their effects
2	D1.3.3–5	Explain the HTT and CCR5 examples, outline what causes mutations, and explain what random means
3	D1.3.6–7	Compare mutations in gamete-forming cells with those in body cells, and explain their role in variation

What is a gene mutation?

original gene



one base changed



A **gene mutation** is a change to the **base sequence** of a gene.

Because the base sequence sets the amino acid sequence, a mutation may change the protein, or it may not.

Not the same as

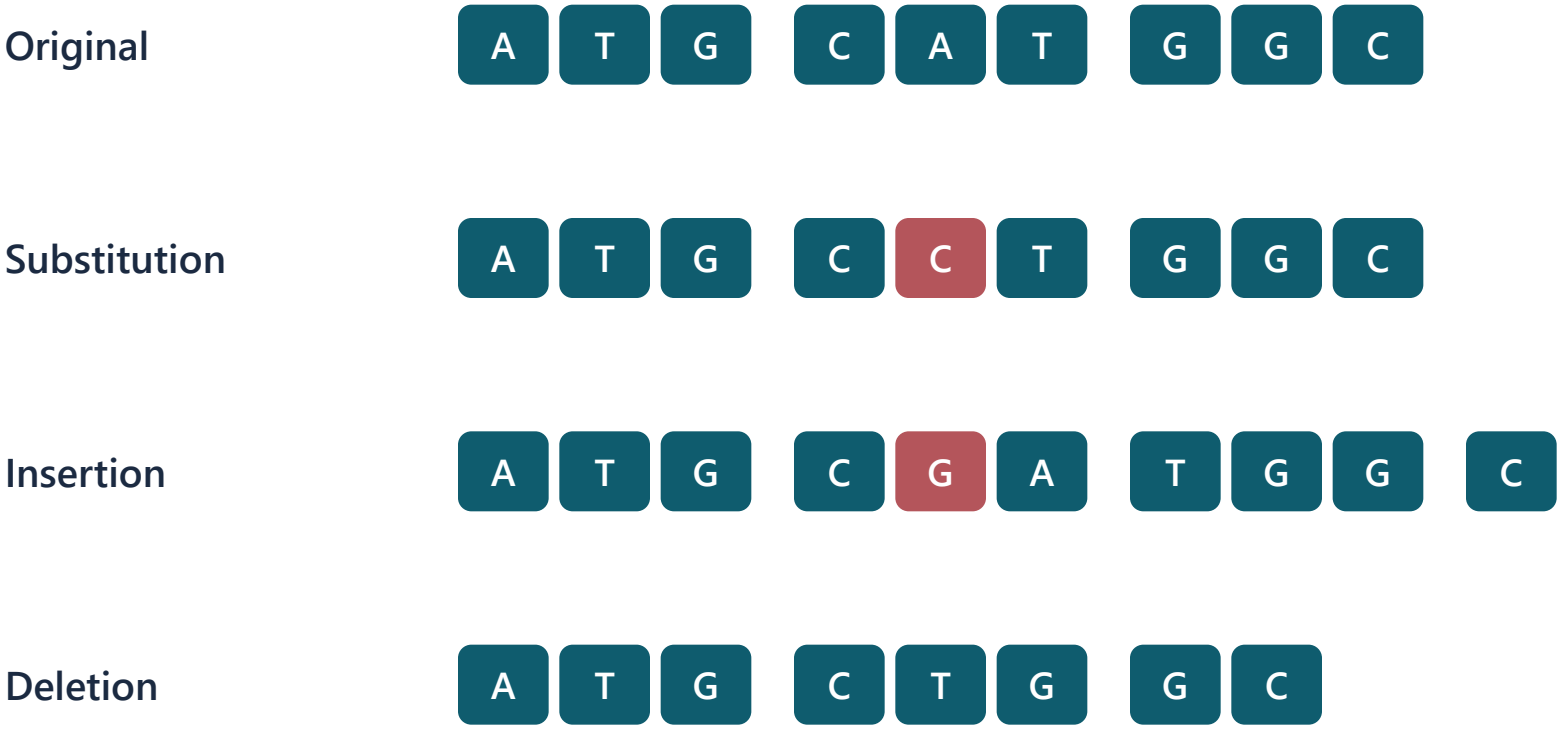
A change in chromosome number (D2.1), or a change in an amino acid, which may or may not follow.

Question for today

Why do some mutations do nothing while others cause disease?

Three types of gene mutation

D1.3.1



Substitution: one base replaced.

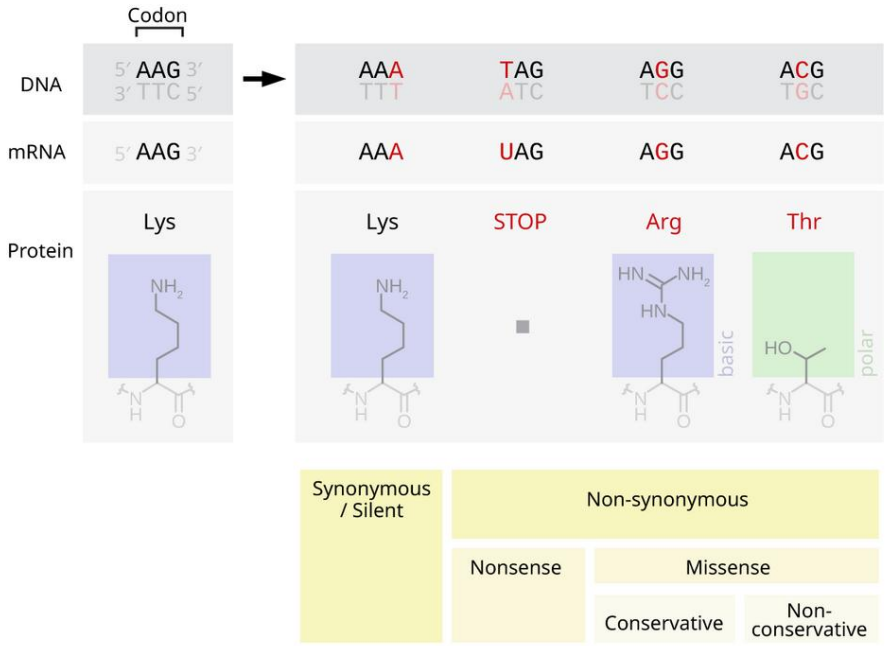
Insertion: a base added.

Deletion: a base lost (here the A of CAT).

Drawn sequences, read in codons of three. Red: the changed base.

What a base substitution can do

D1.3.2



Effect	Example	Result
Silent	GCU to GCC	Same amino acid: the code is degenerate
Missense	GAG to GUG	One amino acid changes
Nonsense	UAC to UAA	Stop codon: polypeptide ends early

Point mutations and their effects. Jonsta247, CC BY-SA 4.0.

A **single-nucleotide polymorphism (SNP)** is a position where one base differs between individuals: the result of a substitution.

Example

Sickle cell anaemia (D1.2) is a missense substitution.

Insertions and deletions: frameshifts

D1.3.3

	mRNA codons	Amino acids
Original	AUG CAU GGC UUA	Met – His – Gly – Leu
One base deleted	AUG CUG GCU UA...	Met – Leu – Ala ...
Three bases deleted	AUG GGC UUA	Met – Gly – Leu

Bases are read in threes, so adding or losing **one** base shifts every codon after it: a **frameshift**. The protein usually stops working.

Adding or losing **three** bases keeps the frame.

Try it

Delete the first base of the second codon in AUG CAU GGC. Write the new codons.

Practice: name the mutation

D1.3.1–3

	Original mRNA	Mutated mRNA
P	AUG UUU GGC	AUG UUC GGC
Q	AUG UUU GGC	AUG UUG GGC A
R	AUG UUU GGC	AUG UAA GGC
S	AUG UUU GGC	AUG GGC

1. Name each type of mutation.
2. Predict the effect of each on the polypeptide.

Working with data: where substitutions fall

D1.3.2

Substitutions found in one gene in a population. The values are invented for practice; the pattern is realistic.

Effect	Number found
Silent	46
Missense	38
Nonsense	6
Total	90

1. Calculate the percentage that are silent.
2. Explain why so many are silent.
3. Suggest why nonsense mutations are rarest in a population.

Check-in: class 1

D1.3.1–3

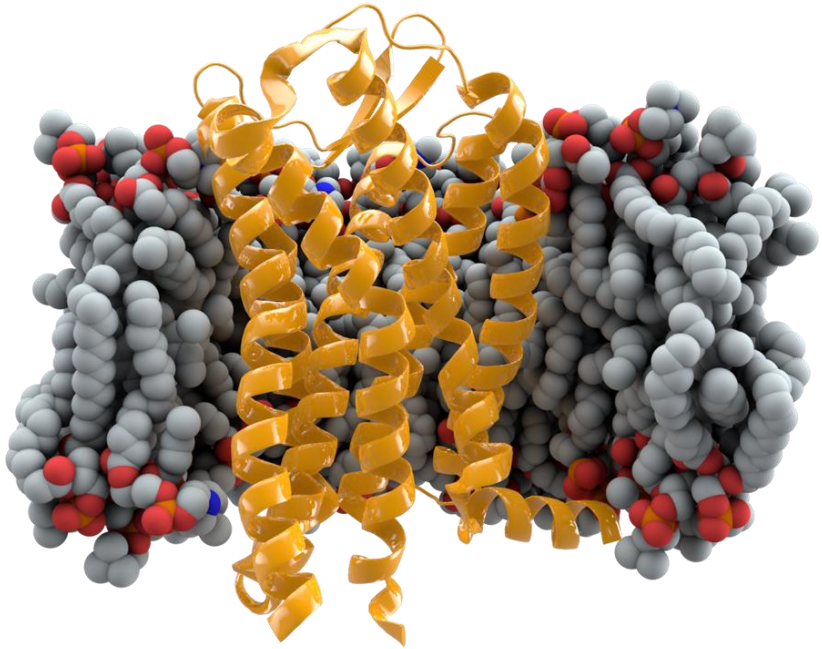
1. Define gene mutation.
2. Name the three types of gene mutation.
3. Explain why a substitution may have no effect.
4. Explain what a frameshift is.
5. Define a SNP.

How to use this

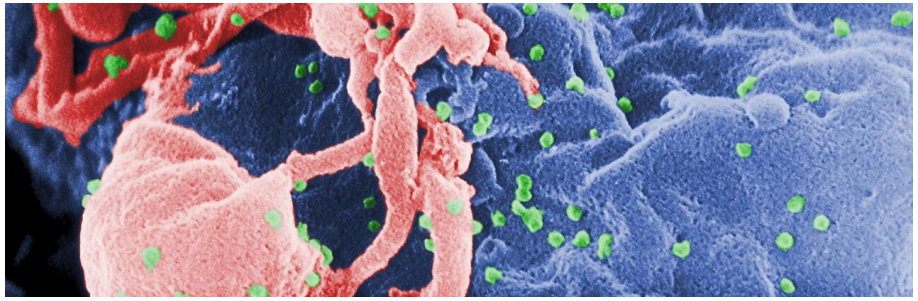
Answer without your notes, then check against slides 3–8.

A deletion: delta 32 in the CCR5 gene

D1.3.3



CCR5 in a cell membrane. Thomas Splettstoesser, CC BY-SA 3.0.



HIV budding from a cell. C. Goldsmith, public domain.

CCR5 is a receptor on some white blood cells. Most HIV strains use it to get in.

Delta 32 deletes 32 bases: a frameshift, so no working receptor reaches the surface. People with two copies strongly resist these HIV strains.

Key idea

A rare case where losing a working protein is an advantage.

An insertion: repeats in the HTT gene

D1.3.3



CAG repeats	Effect	Why
Up to about 35	Typical protein	Normal range
40 or more	Huntington's disease	Long chain of glutamines

Not a frameshift

Each extra repeat is three bases, so the frame is kept. The protein is longer and damages neurons, usually from middle age.

Working with data: CAG repeats

D1.3.3

Number of CAG repeats in HTT and the effect (approximate published ranges).

CAG repeats	Effect
Under 27	Normal
27 to 35	Normal, may expand in children
36 to 39	May or may not develop the disease
40 or more	Will develop it; more repeats, earlier onset

1. State the type of mutation.
2. Explain why the protein is longer.
3. Suggest why repeats can increase between generations.
4. Discuss whether people should be tested.

What causes gene mutations





Tobacco smoke (C. E. Vipin, CC BY-SA 2.0); radiation symbol (public domain).

Cause	Examples
Chemical mutagens	Tobacco smoke, benzene, asbestos
Radiation	Short-wave UV, X-rays, gamma rays
Copying errors	DNA polymerase adds a wrong base
Repair errors	Damage mended with the wrong sequence

Key idea

A **mutagen** raises the mutation rate. Radiation is mutagenic when it has enough energy to change DNA chemically.

Mutation is random

D1.3.5

Mutations can happen **anywhere** in the genome. Some bases mutate more often, so random does not mean equally likely.

No natural mechanism changes a particular base to produce a useful trait.

Common mistake

Organisms do not mutate because they need to. Antibiotics do not cause resistance mutations; they kill bacteria that lack them.

Think

Most mutations land in DNA that codes for nothing. Why are many mutations never noticed?

Practice: true or false?

Sunburn

UV in sunlight can change the base sequence in skin cells.

Need

Cold makes bears mutate to grow thicker fur.

Smoking

Chemicals in smoke raise the mutation rate in lung cells.

For each one

Say whether it is true or false, and explain using the words mutagen and random.

Check-in: class 2

D1.3.3–5

1. Explain how delta 32 gives resistance to HIV.
2. Explain why an HTT repeat expansion is not a frameshift.
3. Give two chemical mutagens and two forms of mutagenic radiation.
4. Define a mutagen.
5. Explain what is meant by saying mutation is random.

How to use this

Answer without your notes, then check against slides 10–15.

Germ cells and somatic cells

D1.3.6

Mutation in a germ cell (makes gametes)

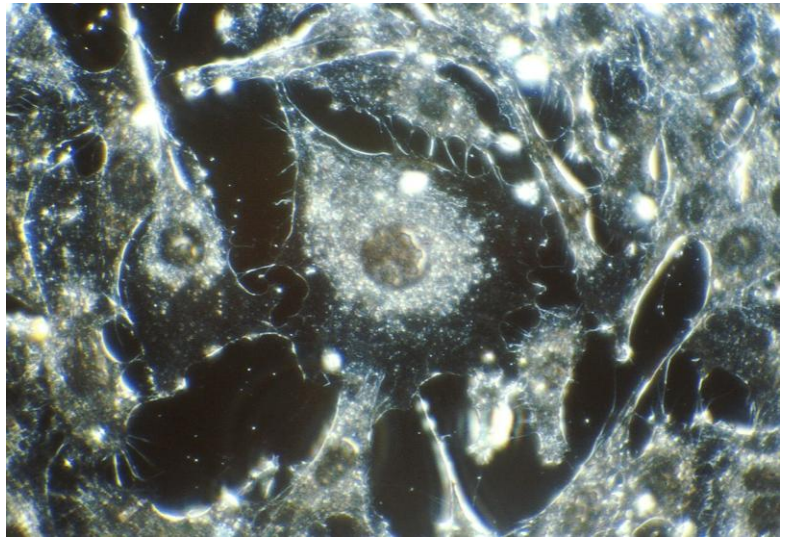
Can be passed to offspring, and then may be in every cell of the child.

Mutation in a somatic cell

Only in cells descended from that one cell. Not inherited. Can lead to cancer.

Cancer

A tumour forms when mutations in genes controlling cell division cause uncontrolled division. Several mutations are usually needed, so risk rises with age and exposure to mutagens.



Cancer cells. Dr. Cecil Fox, public domain.

The source of all genetic variation

D1.3.7



Peppered moths: colour forms arose by mutation. Chiswick Chap, CC BY-SA 3.0.

New **alleles** arise only by mutation. Meiosis and fertilization mix alleles but never make new ones.

Most mutations are neutral or harmful, but without them there would be nothing for natural selection to act on.

Key idea

Harmful to many individuals; essential for a species over long periods (D4.1).

Useful mutations

D1.3.7



A barley field. Public domain.

Plant breeders have used radiation to cause mutations in seeds and then selected useful ones: shorter stems, earlier ripening.

Many barley varieties used in brewing come from such mutants.

Think

Why must breeders screen thousands of plants to find one useful mutation?

Practice: will it be inherited?

D1.3.6

Skin cell

A UV-induced mutation in a skin cell of a parent.

Sperm

A mutation in a cell in the testis that forms sperm.

Lung cell

A smoking-related mutation in a lung cell.

For each one

Decide whether the mutation could be passed to children, and state one possible effect on the person.

Working with data: cancer and age

D1.3.6

New cancer cases per 100 000 people per year. The values are invented for practice; the pattern is realistic.

Age / years	Cases
20–29	40
40–49	180
60–69	900
80+	2200

1. Describe the trend.
2. Explain it using the idea that several mutations are needed.
3. Suggest one way people can reduce their risk.

Check-in: class 3

D1.3.6–7

1. Distinguish between germ cells and somatic cells.
2. State which mutations can be inherited.
3. Explain why cancer risk rises with age.
4. Explain why mutation is the source of all genetic variation.
5. Explain why most mutations are not useful.

How to use this

Answer without your notes, then check against slides 17–21.

Common mistakes to avoid

Not this	But this
All mutations are harmful.	Most are neutral; a few harmful, a few beneficial.
Mutations happen because an organism needs them.	They are random; selection acts afterwards.
Deleting three bases causes a frameshift.	It removes one codon; the frame is kept.
Sunburn mutations are passed to children.	Only germ cell mutations are inherited.

Summary: mutation

D1.3.1–7

	Key idea
D1.3.1	Substitution, insertion, deletion change the base sequence
D1.3.2	Substitutions: silent, missense or nonsense; SNPs
D1.3.3	Frameshifts; CCR5 delta 32; HTT repeats
D1.3.4	Mutagens: chemicals and radiation; copying errors
D1.3.5	Random: anywhere, not directed by need
D1.3.6	Germ cell mutations inherited; somatic ones can cause cancer
D1.3.7	Mutation is the only source of new alleles

Check that you can...

1. **Distinguish** between substitutions, insertions and deletions
2. **Explain** the effects of substitutions and frameshifts, with examples
3. **Outline** the causes of gene mutation
4. **Explain** why mutation is random
5. **Compare** mutations in gamete-forming cells and in body cells
6. **Explain** mutation as the source of genetic variation

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Explain why a substitution may have no effect on a polypeptide.
2. Explain why deleting one base usually does more harm than substituting one base.
3. State one chemical mutagen and one form of mutagenic radiation.

Image credits

Barley field.jpg – Public Domain Images – Public domain

Point mutations-en.png – Jonsta247 – CC BY-SA 4.0

CCR5+membrane1.png – Thomas Splettstoesser (www.scistyle.com) – CC BY-SA 3.0

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